Sensory Innervation of Ocular Tissues

Images used: 13.1, 13.4, 13.3, 10.18, 13.5, 13.6, 13.7

I. Sensory innervation: Trigeminal nerve (cranial nerve 5)

A. Ophthalmic division of the trigeminal nerve

1. Nasociliary nerve is formed by the combination of 5 branches.

a. Infratrochlear nerve

(1) Innervation of

b. Anterior ethmoid nerve

(1) Innervation

(2) Clinical correlate: herpes zoster

i. Hutchinson sign:

c. Posterior ethmoid nerve

(1) Innervation of

d. Long ciliary nerves

(1) Innervation of

e. Short ciliary nerves

(1) Innervation

(2) Pathway

i. Pass through the ciliary ganglion as the sensory root of the ciliary ganglion

2. Frontal nerve

    a. Supratrochlear nerve

        (1) Innervation:

    b. Supraorbital nerve

        (1) Innervation:

3. Lacrimal nerve

    a. Innervation:

4. The nasociliary, frontal, and lacrimal nerves join to form the ophthalmic division of the trigeminal nerve.

B. Maxillary division of the trigeminal nerve

1. Infraorbital nerve

    a. Innervation

2. Zygomatic nerve

    a. Innervation:

Autonomic Innervation to Ocular Tissues

Figures used: 14.2, 14.3

I. Sympathetic pathway to the internal ocular structures

A. Central pathway

B. Preganglionic fibers

C. Postganglionic fibers

D. Innervation

1. Iris dilator:
2. Choroidal vasculature:
3. Muller muscle:

II. Parasympathetic pathway to the internal ocular structures

A. Central pathway

B. Preganglionic fibers

C. Postganglionic fibers

D. Innervation

1. Iris sphincter:
2. Ciliary muscles:

III. Autonomic pathway to the lacrimal gland

A. Pathway

B. Innervation

1. Parasympathetic:
2. Sympathetic:

C. Clinical correlate: corneal reflex

Skeletal Motor Innervation to Ocular Tissues

Images used: 13.10, 13.7, 10.18, 13.11, 13.12, 13.13, 13.14, 13.16, 13.17, 13.18

I. Oculomotor nerve (cranial nerve 3)

A. Innervates

B. Oculomotor nucleus:

C. Cranial nerve 3 pathway

1. Leaves the brainstem at the level of the midbrain.

2. Passes between

3. Follows the

a. Clinical correlate: CN3 is most frequently affected by posterior communication artery aneurysm but can be affected by any of the surrounding arteries.

4. Cavernous sinus

5. Superior orbital fissure

6. Orbit

    a. Superior branch: innervates the SR and levator muscle

    b. Inferior branch

D. Clinical correlate: cranial nerve 3 palsy

    a. Ptosis

    b. The eye is positioned down and out due to the unopposed actions of the SO and LR

    c. They cannot adduct (MR)

    d. In the abducted position, the eye cannot move up or down (SR, IR)

    e. In the adducted position, the eye cannot move the eye up (IO)

    f. Pupil dilation

    g. Decreased accommodation

II. Trochlear nerve (cranial nerve 4)

A. Innervates:

B. Nucleus:

C. Cranial nerve 4 pathway

1. Leaves the brainstem from the posterior midbrain and then crosses to the opposite side.

2. Follows the superior cerebellar and posterior cerebral arteries around the brainstem

3. Cavernous sinus

4. Superior orbital fissure

5. Orbit

D. Clinical correlate: cranial nerve 4 palsy

1. Eye deviation in primary position:

2. Restrictions:

III. Abducens nerve (cranial nerve 6)

A. Innervates:

B. Nucleus:

C. Cranial nerve 6 pathway

1. Exits the groove between the pons and medulla

2. Runs along the occipital bone and posterior slope of the petrous portion of the temporal bone

    a. Makes a sharp bend over the petrous ridge

    b. Clinical correlate: Susceptible to trauma and increased intracranial pressure.

3. Cavernous sinus

4. Superior orbital fissure

5. Orbit

D. Clinical correlate: CN 6 palsy

1. Eye deviation in primary position:

2. Restrictions: